ANGKOR ARCHITECT

Detailed Design Document & Technical Architecture

Genre: Isometric Simulation / City Builder **Platform:** Web (HTML5 WebGL / PWA) **Target Date:** Q2 2027

1. Executive Summary & Vision

Angkor Architect is a web-based, isometric resource management and city-building game inspired by the engineering marvels and cultural legacy of the medieval Khmer Empire (9th–15th century). Players assume the role of the **Royal Master Builder (*Kru Vitu*)** appointed by the Sovereign.

The primary challenge revolves around designing, constructing, and maintaining the vast hydraulic grid (*Barays* and interconnected canals) that powered ancient Angkor, while directing human labor, quarrying sandstone from Mount Kulên, and raising monumental temple complexes to earn divine favor (*Bun*).

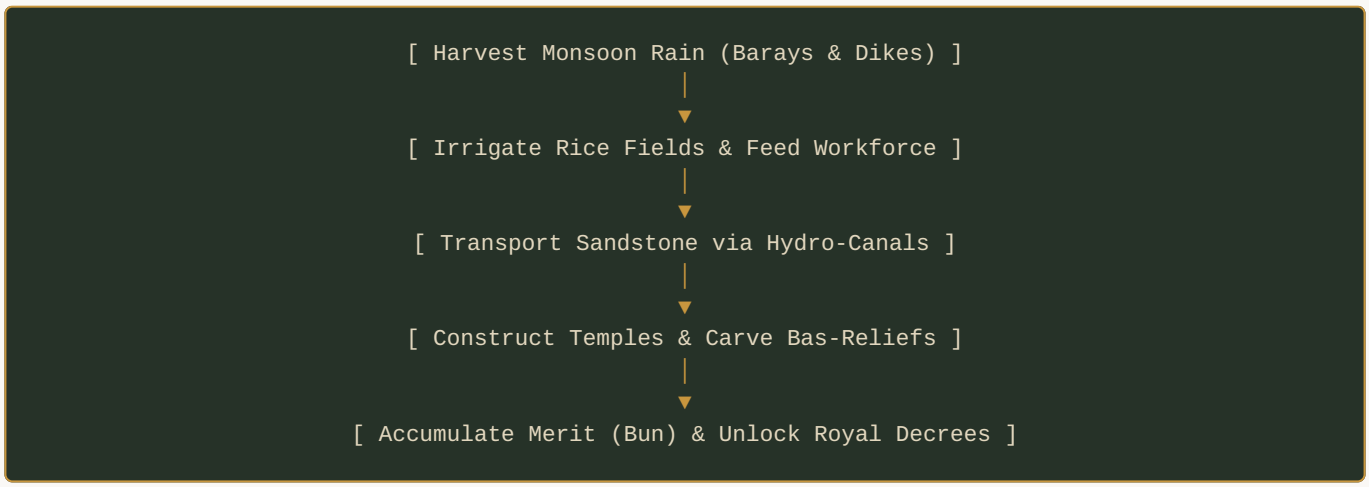
> **Core Philosophy: Water as Life & Threat**
>
> *Unlike modern city builders where road transport dominates, in Angkor Architect, **waterways are the primary lifeblood**. Water dictates agricultural yield, powers logistics via rafts, stabilizes deep stone foundations, and presents constant seasonal flood risks.*

2. Historical Setting & World Design

The game takes place across the peak centuries of the Khmer Empire, encompassing three distinct ecological sub-regions:

- **Tonlé Sap Floodplain:** Highly fertile lacustrine zone subject to massive seasonal water shifts. Requires robust floodgates and elevated silt management.
- **Kulên Foothills:** Dense forest and sandstone-rich mountainous terrain requiring elevated canal locks and steep haulage routes.
- **Angkor Core Region:** Flat clay-and-sand terrain optimal for massive rectangular city grids, Moat-surrounded enclosures, and grand temple terraces.

3. Core Gameplay Loop & Systems



A. The Hydraulic Engine (Monsoons & Water Flow)

The game engine runs a continuous 2-phase seasonal cycle:

- **Wet Season (Monsoon):** Torrential rain refills rivers and reservoirs. Players must balance floodgate release to prevent dikes from breaking and wiping out rice paddies.
- **Dry Season:** Water levels drop severely. Agricultural output depends entirely on the volume stored in *Barays* during the wet months.

B. Resource Logistics Matrix

Resource	Primary Source	Function / Mechanics	Transport Requirement
Water	Monsoon / Rivers	Crop growth, canal transport, foundation stability	Canals, Spillways, Floodgates
Rice & Fish	Paddies & Tonlé Sap	Sustains artisans, quarry workers, and monks	Roads & Shallow Boats
Sandstone	Kulên Quarries	Primary material for Temple Prasats & Wall Reliefs	Heavy River Barges (Canals)
Laterite	Underground Digs	Sub-surface foundational blocks & city boundary walls	Ox Carts & Foot Workers
Merit (Bun)	Active Temples	Meta-currency for technology tree & Royal Decrees	Global Value Accumulation

C. Temple Construction Pipeline

Constructing a monument like Angkor Wat requires four distinct sequential engineering steps:

1. **Moat & Water Table Stabilization:** Excavating wide surrounding moats to keep the subsoil moist, preventing structural collapse under heavy stone weight.
2. **Laterite Foundation:** Laying heavy, porous laterite blocks as a solid structural base.
3. **Sandstone Superstructure:** Stacking fine-grained sandstone tiers and raising central *Prasat* (Lotus-bud towers).
4. **Bas-Relief Sculpting:** Mini-game phase where players select historical, mythological (e.g., *Reamker*), or daily life motifs, granting cultural buffs and boosting civic morale.

4. Technical Web Architecture

Angkor Architect is engineered for instant, zero-install browser play across desktop and tablet devices.

- RENDER ENGINE **PixiJS / WebGL:** Delivers 60 FPS performance on 2D isometric grids with multi-layered sprite batching.
- LANGUAGE **TypeScript:** Ensures strict data types across simulation grids, water flow vectors, and game state transitions.
- HYDRAULICS LOGIC **Cellular Automata:** Custom grid-based flow algorithm that calculates water pressure, elevation differentials, and volume propagation across canal nodes every tick.
- PATHFINDING **A* Algorithm:** Optimizes worker navigation and boat routing along active waterways.
- STORAGE **IndexedDB:** Local save state persistence with compressed JSON strings, paired with lightweight Cloud sync (Supabase).

5. Development Roadmap

Phase	Timeline	Key Deliverables
Phase 1: Prototype	Months 1 – 2	Isometric grid rendering engine, tile mapping, basic water flow logic, and stone/rice resource loop.
Phase 2: Alpha	Months 3 – 4	Dynamic seasonal system, full hydraulic building set (Barays, gates), construction pipeline, and UI layout.
Phase 3: Beta	Month 5	Soundtrack integration (Pinpeat instruments), Bas-relief mini-game, Campaign Level 1 (Roluos Group).
Phase 4: Launch	Month 6	PWA mobile optimization, performance profiling, localizations, and public release.

6. Key Differentiators

1. **Unexplored Setting:** First major web strategy game focused explicitly on Khmer hydro-engineering and architectural heritage.
2. **Realistic Fluid Simulation:** Transcends simple binary utility toggles in favor of dynamic seasonal water dynamics.
3. **Cultural Authenticity:** Developed with close adherence to historical Khmer architectural layouts, art motifs, and musical soundscapes.